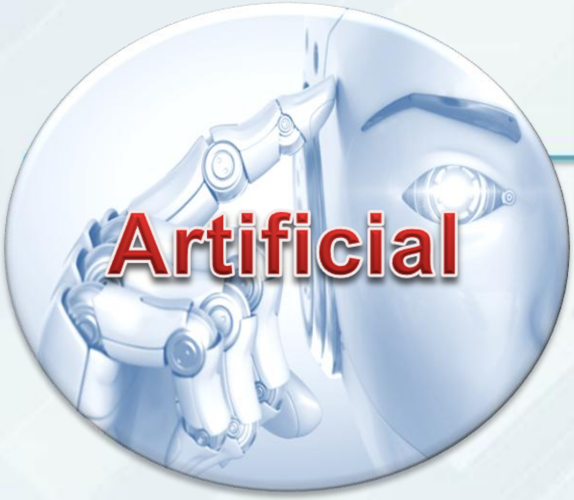
MODULE 1: FOUNDATIONS & PHILOSOPHY OF AI

Introduction

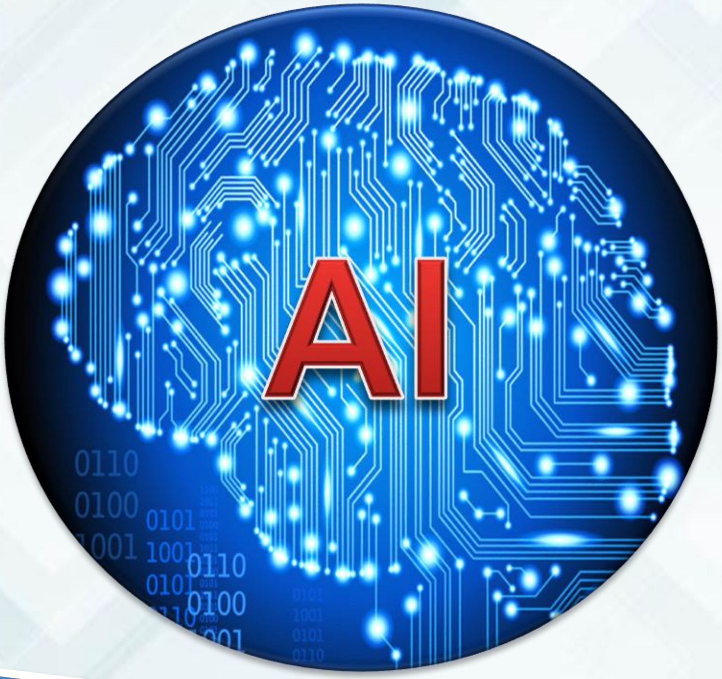
- AI has always intrigued people.
- For example, conversing with a computer NLP.
- Science fiction has long anticipated the rise of machine intelligence.
- Today, a new generation of self-learning computers is reshaping every aspect of our lives.

OUTLINE

- Introduction
- Definition
- History / evolution
- Branches (Sub field) of AI
- Classification/ types
  - Narrow AI
  - Artificial General Intelligence
  - Artificial Super Intelligence
- Applications, examples
- Languages : Python, Java, Lisp



Artificial is something
that is made or created by humans,
not natural.



The ability to learn, understand, reason,
and solve problems

Definition

- AI is simply a human-made systems that are designed to perform tasks that normally require human intelligence.
- It is a wide field of new types of computers that are trained rather than programmed
- AI is concerned with intelligent behaviour in artifacts.

4 Goals of AI

1. Thinking Humanly (Cognitive Science Approach)
2. Thinking Rationally (The “Laws of Thought” Approach)
3. Acting Humanly (The Turing Test Approach)
4. Acting Rationally (The Rational Agent Approach)

Thinking Humanly

This the ability to make computer systems think (i.e. machines with minds).
It deals with the automation of human thinking with regards to activities such as; decision making, problem solving, learning, etc.

Thinking Rationally

This deals with the study of mental faculties through the use of computational models. Computations that makes it possible for machines to perceive, reason, and act.

Acting Humanly

This is simply the act of creating AI systems that perform functions that requires intelligence when performed by humans.

An example here can be an NLP AI systems.

Acting Rationally

This deals with designing AI systems that can choose and perform the best possible action to achieve a goal based on the information available to them.

Rather than focusing on only how machines thinks, it focuses on a machine acts to provide the most effective outcome.

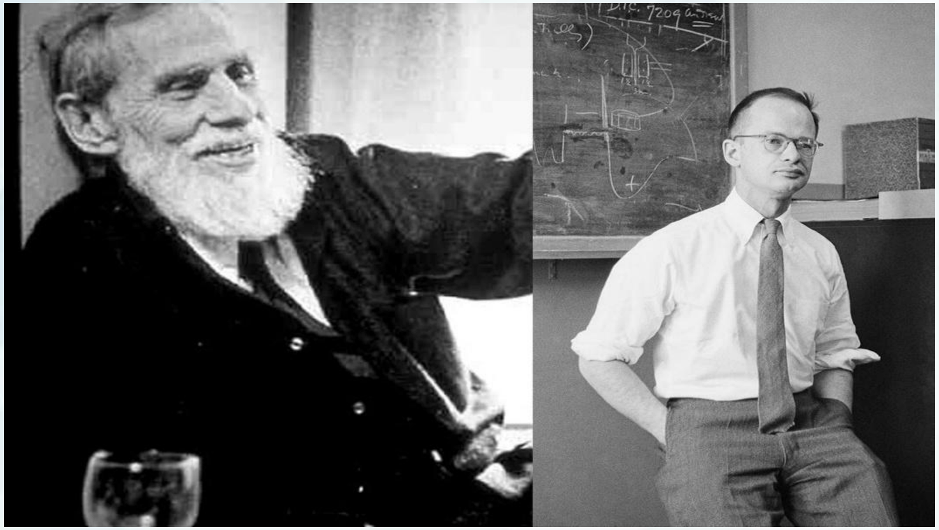
AI Techniques

Instead of relying solely on fixed rules, AI systems can learn from data, identify patterns, make decisions, and improve their performance over time, Unlike traditional programming which relies on algorithms to attained a solution.

History of AI

1943:Foundation of Neural Network established by Warren McCulloch and Walter Pitts

Drawing parallels between the brain and computing machines



History of AI

In 1950, Alan Turing introduces the test-

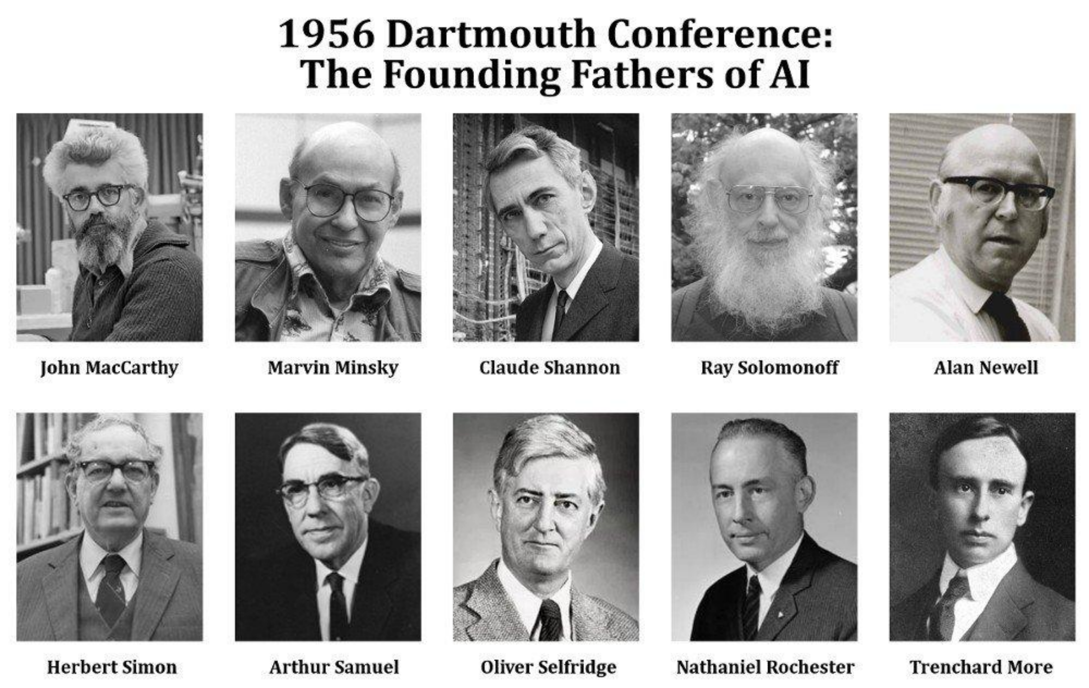
A way of testing a machine's intelligence

Computers those indistinguishable from human beings

Turing predicted intelligent computing in 50 years

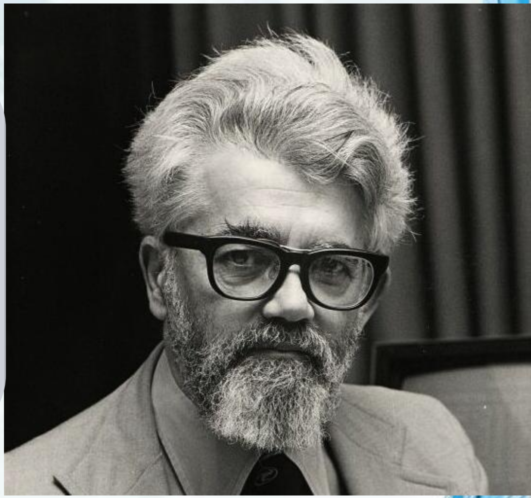
History of AI

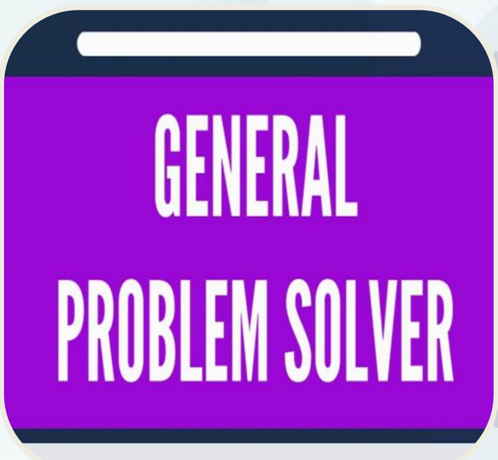
The term **Artificial Intelligence** was first coined by John McCarthy in Dartmouth conference devoted to the topic in 1956



History of AI

- 1958 In MIT lab, McCarthy defined the high level language Lisp, a dominant AI programming language





- 1961 Newell and Simon's presented the GPS, a program to imitate human problem-solving protocols

History of AI

1966 ELIZA a natural language program created by Joseph Weizenbaum at MIT

ELIZA handles dialogue on any topic, similar in concept to todays chatbot

History of AI

Dendral is considered the first expert system

Developed at Stanford University by Edward Feigenbaum, Bruce G. Buchanan, Joshua Lederberg

It automated the decision-making process and problem-solving behavior of organic chemists.

Used to identify the structure of chemical compounds

History of AI

1970 MYCIN expert system in medical domain

Developed at Stanford University by Edward Shortliffe

Identify bacteria causing severe infections, and to recommend antibiotics, diagnosis of blood clotting diseases

It was written in Lisp

History of AI

Edward Feigenbaum, "father of expert systems“

1980-1985 proliferation of expert systems

History of AI

- AI winter 1985-95
- Emergence of Intelligent agents
- 1997 Computer Deep Blue beats world chess champion Gary Kasparov
- Availability of large data sets, distributive computing, Explosion of Internet
- Processing power, Proliferation of GPU
- 2002 Irobot Roomba autonomous vacuum cleaner
- ASIMO humanoid robot created by Honda in 2000
- 2009 Google build s first self driving car

History of AI

2011 : IBM,s Watson defeats champions of US game show Jeopardy

2011-14 : Personal assistants like Siri, Google Now, Cortana use speech recognition to answer questions and perform simple tasks

2016 : ALphaGO beats professional Go player Lee Sedol

Artificial intelligence systems take on more tasks and solve more problems

Types of Artificial Intelligence

Generally, A.I. falls within three categories:

- **Artificial Narrow Intelligence (ANI)**
- **Artificial General Intelligence (AGI)**
- **Artificial Super Intelligence (ASI)**

Narrow/Weak AI

- AI trained to perform a single, specific task (e.g., facial recognition, spam filters, recommendation engines).

- Narrow AI can identify pattern and correlations from data more efficiently than humans. But it cannot expand and take tasks beyond its field

Artificial General Intelligence

General/Strong AI

- AI systems with generalized cognitive abilities which find solutions to the unfamiliar task it comes across.
- ability to understand context and make judgments based on it
- Over time, it learns from experience, is able to make decisions even in times of uncertainty or with no prior available data, use reason, and be creative.
- Intellectually, these computers operate much like the human brain.
- So far we've not been able to do it, although most believe we might be able to do so sometime in future

Artificial Super Intelligence

- Artificial Super Intelligence (ASI) refers to the position where computer/machines will surpass humans and machines would be able to mimic human thoughts.

- It refers to a situation where the cognitive ability of machines will be superior to humans.

- AI robots would be able to think for themselves, attain consciousness, and operate without any human involvement, perhaps at the direction of another A.I

Artificial Super Intelligence

- There is still no machine that can process the depth of knowledge and cognitive ability as that of a fully developed human.
- ASI had two school of thoughts,
  1. On one side great scientist like Stephen Hawking saw the full development of AI as a danger to humanity
  2. Demis Hassabis, Co-Founder & CEO of DeepMind believes that the smarter AI becomes the better the world would be and a helping hand to mankind.

Most Popular AI Languages

- Python
- Lisp
- Prolog
- Java
- C++

Branches of AI

- Expert Systems
- Machine Learning
  - Artificial Neural Network (Deep Learning)
- Natural Language Processing
  - Speech Recognition
- Robotics
- Computer Vision
  - Image Recognition

Expert Systems

- An expert system is a computer system that emulates, or acts in all respect, with the decision making capabilities of a human expert (E.g. Medical Diagnosis expert system)
- Knowledge Based System
- Expert to have extensive knowledge

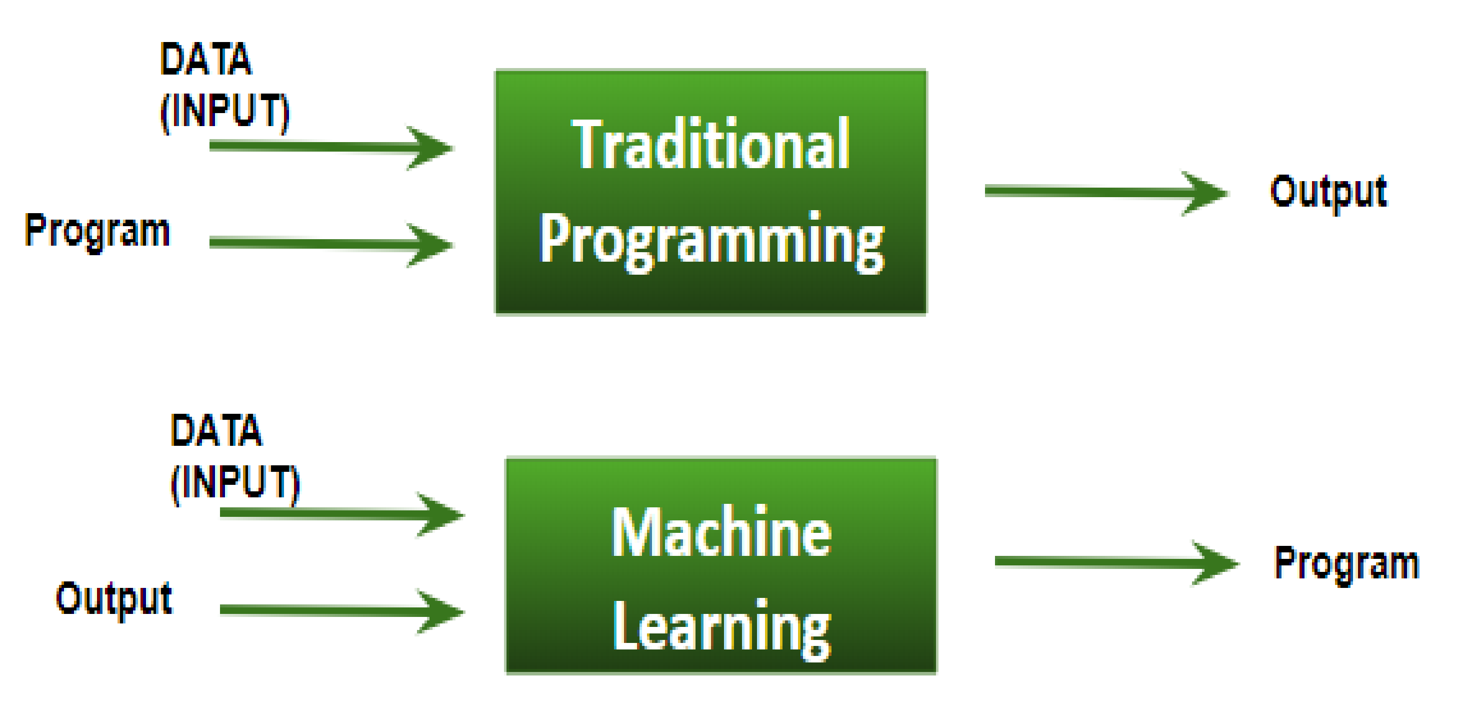
Machine Learning

- **Machine Learning** is the field of study that gives computers the capability to learn without being explicitly programmed.

  Arthur Lee Samuel

- More similar to humans: ***The ability to learn***.
- Machine learning is about predicting the future based on the past.
- The process starts with feeding good quality data and then training machines(computers) by building machine learning models using the data and different algorithms.

Machine Learning



Machine Learning

- **Supervised learning:** Is a type of machine learning where the model is trained using labeled data (data with correct answers). This system learns by example and is then able to predict outcomes for new data.

- **Unsupervised Learning:** is type of learning where the model is given unlabeled data and must find patterns or groupings on its own.

- **Reinforcement Learning:** is a type of ML where an AI agent learns by interacting with an environment and receiving rewards or penalties for its actions. (e.g. A robot learning to walks, it gets rewards when it moves forward and penalty otherwise.)

Artificial neural networks

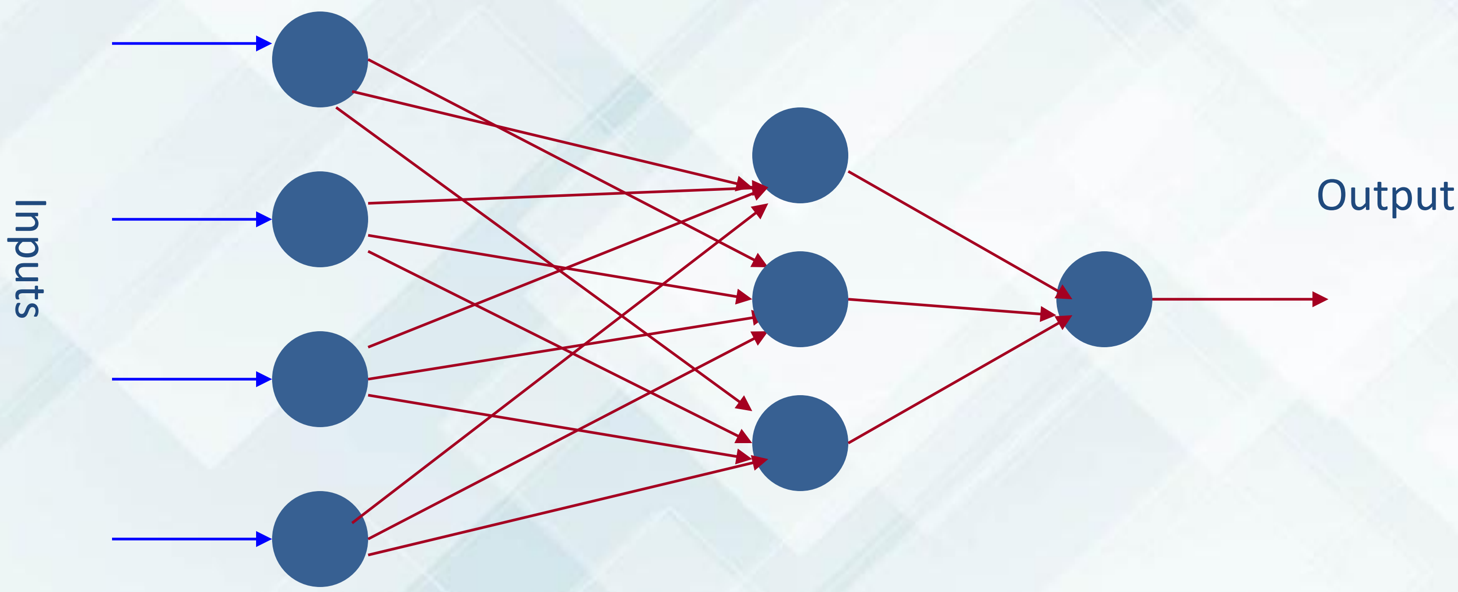
- ..a computing system made up of a number of simple, highly interconnected processing elements, which process information by their dynamic state response to external inputs."

Dr. Robert Hecht-Nielsen

Artificial Neural Networks

- ANNs are composed of multiple **nodes**, which imitate biological **neurons** of human brain. The neurons are connected by links and they interact with each other.
- The nodes can take input data and perform simple operations on the data. The result of these operations is passed to other neurons. The output at each node is called its **activation** or **node value.**
- Each link is associated with **weight.** ANNs are capable of learning, which takes place by altering weight values.

Artificial Neural Networks



An artificial neural network is composed of many artificial neurons that are linked together according to a specific network architecture. The objective of the neural network is to transform the inputs into meaningful outputs.

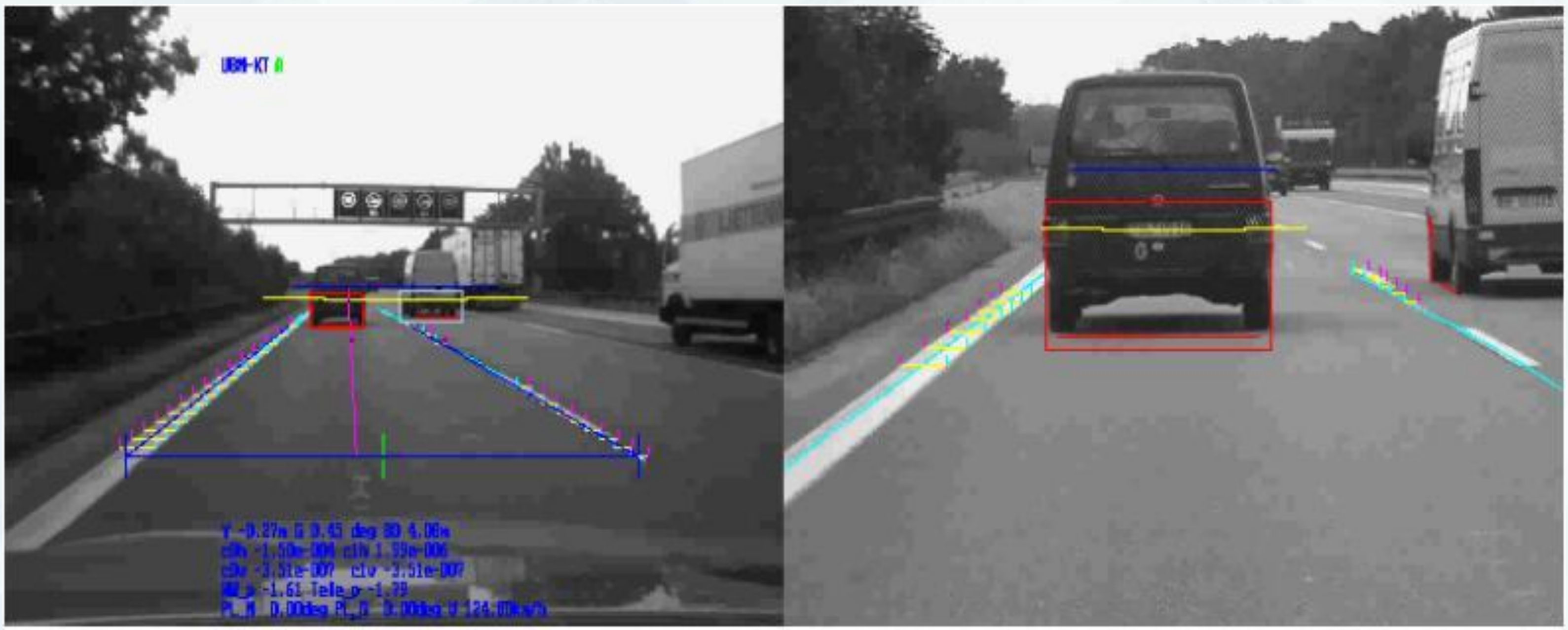
Computer Vision

Is an area of AI that enables computers to understand, interpret, and make decisions from images or videos. Similar to how human vision works. It involves teaching machines to see and extract meaningful information from visual data.

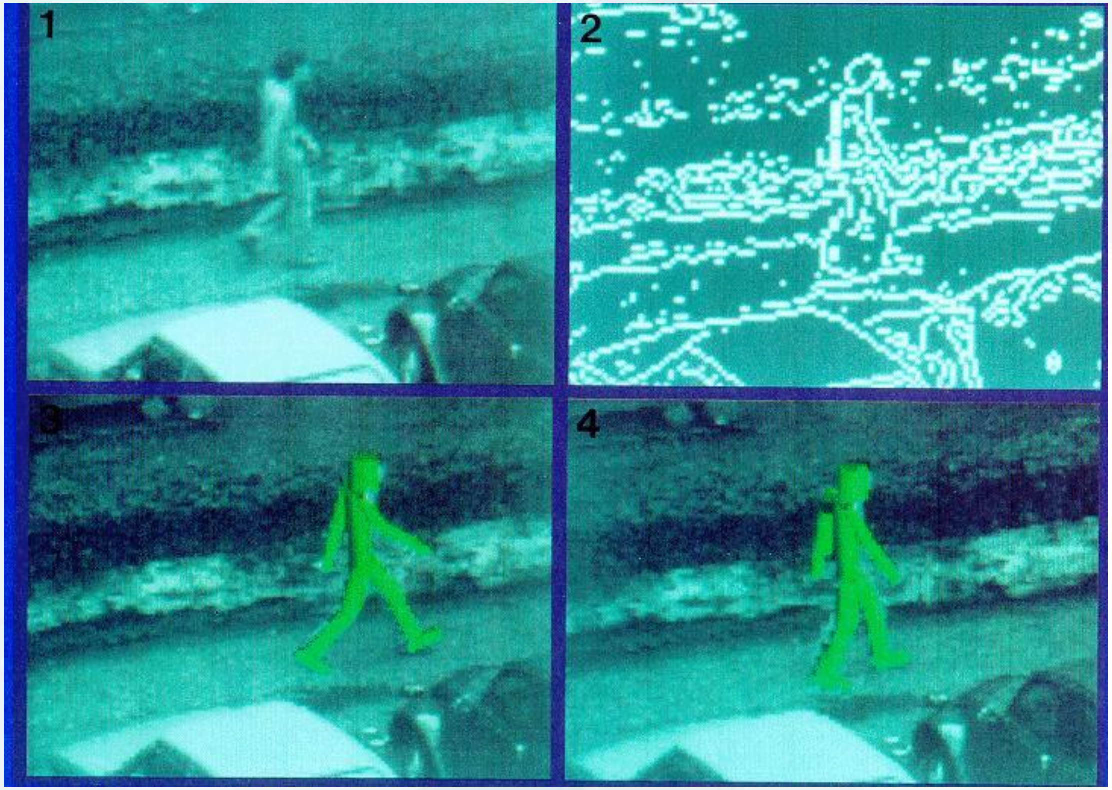
It’s also called:

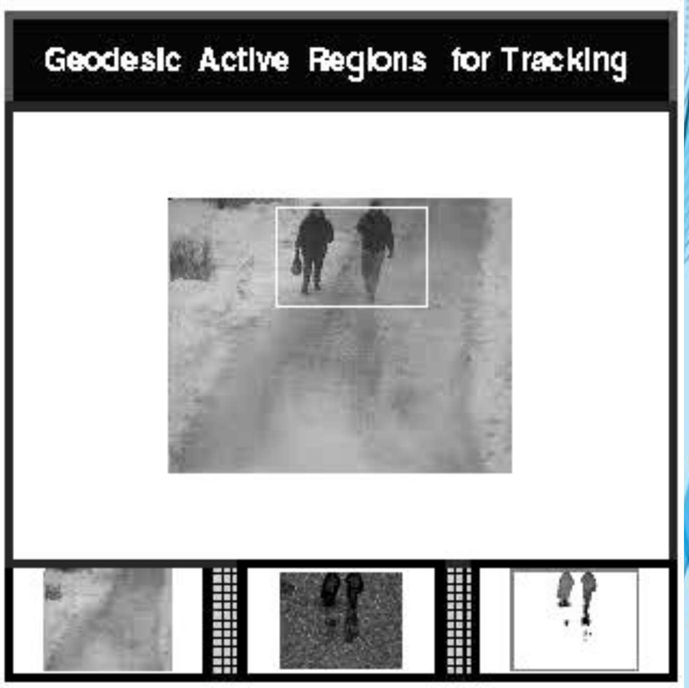
- Image Understanding
- Image Analysis
- Machine Vision

Autonomous Vehicle



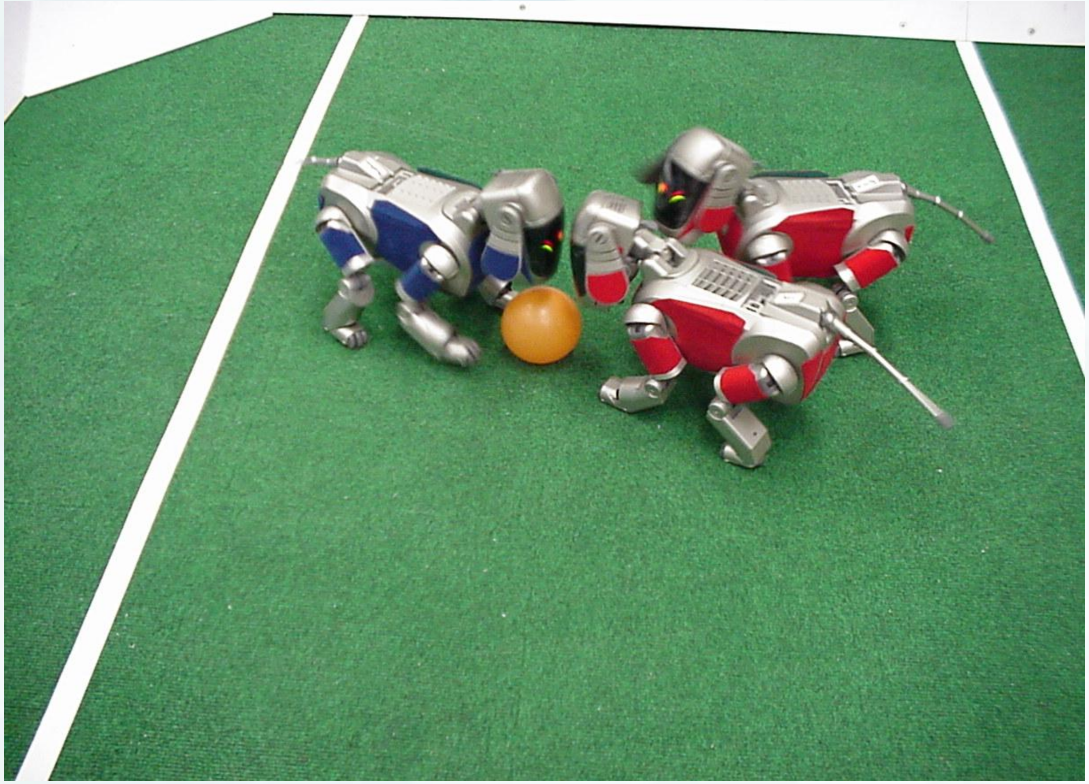
Applications: Surveillance





Robotics

The intersection of AI and mechanical engineering, allowing AI to interact with the physical environment using their sensors and actuators.



Applications of AI

Some of the applications of AI include but not limited to:

- **Healthcare:** Disease prediction, automated radiology, drug discovery.
- **Finance:** Algorithmic trading, fraud detection, credit scoring.
- **Transportation:** Autonomous vehicles (self-driving cars), route optimization.

Advantages and Disadvantages

- High precision
- 24/7 Availability
- Rapid Processing of Big data
- Automation of Dangerous Tasks

Disadvantages:

- High cost of development
- Massive energy requirements
- Lack of True Emotional Intelligence
- Potential job displacement

The Philosophy of Intelligence: Two Thought Experiments

When defining what makes a machine "intelligent," computer scientists and philosophers frequently debate two foundational thought experiments.

The Turing Test (Alan Turing, 1950)

Turing proposed a pragmatic, operational test to bypass the philosophical definition of "thinking."

● **The Setup:** A human evaluator sits in a room and has a text-based, typed conversation with two entities in another room; one is a human, the other is a machine.

● **The Goal:** If the evaluator cannot reliably tell which entity is the machine after a reasonable amount of time, the machine is said to have passed the test.

● **Requirements to Pass:** To succeed, the machine needs:

1. *NLP* to communicate.
2. *Knowledge Representation* to store what it knows.
3. *Automated Reasoning* to use stored information to answer questions.
4. *Machine Learning* to adapt to new circumstances in the conversation.

● *(Note: The* ***Total Turing Test*** *adds video feeds and a physical hatch, requiring the machine to also possess Computer Vision and Robotics).*

The Chinese Room Experiment (John Searle, 1980)

Philosopher John Searle introduced this argument to challenge the concept of Intelligent AI, particularly, the claim that successfully passing the Turing test is sufficient evidence of genuine understanding or the possession of a mind. He brought counter experiment to that of Alan Turing.

- **The Setup:** Imagine John Searle, who speaks *only* English, locked in a room. He is given a massive rulebook written in English. This book contains instructions on how to manipulate Chinese symbols (e.g., "If you see symbol X, write down symbol Y").
- **The Execution:** Native Chinese speakers outside the room slide questions written in Chinese under the door. Searle uses his rulebook to find the matching symbols, writes down the required response symbols, and slides the paper back out.

The Chinese Room Experiment (John Searle, 1980)

● **The Argument:** To the people outside, the room appears perfectly fluent in Chinese; it easily passes the Turing Test. However, Searle *understands absolutely no Chinese*. He is simply manipulating symbols based on their syntax (shape/structure).

● **Significance:** Searle argues that computational models only achieve **syntax** (symbol manipulation), not **semantics** (actual meaning or understanding). Therefore, a computer running a program can only *simulate* intelligence, but it cannot truly *understand*.

● **The Systems Reply (Counter-argument):** Critics argue that while Searle himself doesn't understand Chinese, the *entire system* (Searle + the rulebook + the room) does possess understanding.

Optimal vs. Human-Like Behavior

To conclude, AI can be viewed from different perspectives. But two important approaches are designing AI systems that are ***acting humanly*** and ***acting rationally.***

Optimal vs. Human-Like Behavior

Human-Like Behavior	Optimal Rational Behavior
Focuses on making machines behave like humans.	Focusing on making machines choose the best possible action to achieve a goal
The goal is to imitate human thinking and actions	The goal is to maximize success and efficiency
A system that may make mistakes similar to those made	A system which seeks the effective solution, even if humans would not choose it.
Evaluated Using Test such as the Turing Test	Evaluated based on how well it achieves

Module 2: Intelligent Agents..